MIR MUNAVVAR ALI

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PROFESSIONAL SUMMARY

Mechanical / Mechatronics Engineer with 3+ years of experience across CAD/CAE, robotics, electromechanical design, and hardware prototyping. Experienced in SolidWorks, FEA, actuators, additive manufacturing, BOM/ECO documentation, and MATLAB-based controls, with a focus on design optimization, prototype development, and hardware validation.

EDUCATION

Northeastern University | Boston, MA **Jan 2025 – May 2027**
M.S. Mechanical Engineering, Concentration in Mechatronics
Coursework: Control Systems, Mathematical Methods, Mobile Robotics, Mechatronics Systems, MEMS, Energy Storage & Harvesting

Osmania University | Hyderabad, India **Jul 2018 – Jun 2022**
B.E. Mechanical Engineering
Coursework: CAD/CAM, Machine Design, Strength of Materials, Manufacturing, Electronics, FEA, DFM/DFA, GD&T

INDUSTRY EXPERIENCE

Mechanical Technologist Engineer – Co-op | Veeco Instruments Inc. | Plainview, NY Jan 2026 – Aug 2026
- Optimized neutralizer designs through iterative testing, reducing component size by 2×, developing compact vacuum subassemblies, doubling operational lifetime, and lowering power consumption on a $5M+ ion beam deposition project.
- Prototyped ion beam deposition components using additive manufacturing, including alumina ceramic and semiconductor-grade parts; designed impedance-matching networks and managed drawings, BOMs, and ECOs using SAP/ECTR.

Junior Mechanical Design Engineer | Vir Innovations Pvt. Ltd. | Hyderabad, India Mar 2023 – Dec 2024
- Designed robotics hardware for stair-climbing and vehicle-alerting systems, developing chassis and actuator mounts in SolidWorks and validating servo motors with Arduino; reduced prototyping time by 5× through virtual validation.
- Conducted static and dynamic FEA and validated IP-rated housings through stress testing for corrosion resistance, water resistance, and thermal shielding; supported patent and product deliverables.

Product Design Engineer – Freelance | Freelance | Hyderabad, India Oct 2024 – Dec 2024
- Engineered rugged mechanical housing systems, delivering optimized designs two weeks ahead of schedule through iterative CAD development and client feedback.
- Optimized designs and manufacturing approaches, reducing material waste by 5 kg per product.

Mechanical Design Engineer – Intern | Mowito Automation | Pune, India Nov 2022 – Dec 2022
- Designed a 3-axis Cartesian cobot base and electromechanical actuator in SolidWorks/Onshape; calculated 36.7 N·m leadscrew torque and performed fatigue analysis supporting 100,000+ cycles on a $1M project.
- Integrated hydraulic and pneumatic systems for variable field loads and redesigned seals and housings using additive manufacturing; validated performance through 1.6 kN·m dynamic-load checks.

Mechanical Engineer – Co-op | Hyderabad Batteries Limited | Hyderabad, India Oct 2021 – May 2022
- Designed a 3-axis Cartesian robotic arm for automated pick-and-place applications, using kinematic analysis and cycloidal reducers to achieve 256× resolution and higher torque.
- Streamlined 130 BOM items by consolidating duplicate wiring, connectors, and accessories, reducing component variation by 7% and simplifying procurement and assembly by 10%.

SELECTED PROJECTS

DC Motor Control System | Northeastern University Sep 2025 – Dec 2025
- Built a DC motor speed/position control system using MATLAB/Simulink and Tinkercad with Arduino, H-bridge, potentiometer, and encoder; validated encoder gain at 0.00076 rad/s per count and ±2° positioning accuracy.
- Developed and tuned Simscape motor models and PID controllers, comparing simulation with experimental data to improve control performance.

2D Submarine Control Project | Northeastern University Mar 2025 – Apr 2025
- Linearized nonlinear submarine dynamics, built a state-space model, and assessed controllability and observability; designed state-feedback, observer-based, and multi-axis PID controllers in MATLAB.
- Simulated closed-loop responses to evaluate stability, disturbance rejection, and position-tracking accuracy.

LEADERSHIP & ROBOTICS

Mechanical Analyst / CAE Mechanical Head, Asia Pacific Robot Contest (ABU ROBOCON) | Jun 2019 – Jun 2022
- Led design and deployment of semi-autonomous competition robots, earning 12th- and 15th-place national results; fabricated and assembled hardware using laser cutters, CNC machines, hydraulic presses, and metal-bending equipment.

TECHNICAL SKILLS

CAD / CAE: SolidWorks, Onshape, Fusion 360, AutoCAD, ANSYS, FreeCAD, MeshLab, FEA
Mechanical Design: Machine design, kinematic analysis, actuators, electromechanical systems, DFM/DFA, GD&T, BOMs, ECOs, technical documentation
Controls / Programming: MATLAB, Simulink, Simscape, Python, C, Embedded C, PID, LQR, Arduino
Prototyping / Fabrication: FDM/SLA 3D printing, CNC machining, laser cutting, hydraulic presses, metal bending, soldering, hand and shop tools
Hardware: Motors, sensors, linear actuators, multimeters, pneumatic and hydraulic systems